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EXPERIENCE

Machine Learning Engineer (Founding Team)

May 2026 – Present

Stealth AI startup (consumer product comparison)

Remote

- Sole MLE on a 4-person core team; retained through a 10-to-4 consolidation and converted from intern to full-time, owning all ML, data-infrastructure, and backend decisions ahead of launch.
- Run continued pre-training and multi-task post-training (SFT + RL) of a FLAN-T5 base into a 250M-parameter multi-head specialist covering structured harmonization, vision-based attribute extraction, and recommendation ranking.
- Built the frontier-model teacher pipeline behind training-data generation and quality gating over a 38K-product corpus; the distilled student beats GPT-5.6/Claude/Gemini on a 570-example human-audited benchmark, with schema consistency across product categories that the teacher models do not hold.
- Designed evaluation and release infrastructure: regression gates governing checkpoint promotion over end-to-end cloud GPU training runs (A100/H100).
- Own the production stack: MongoDB Atlas catalog with embedding store and faceted search, FastAPI backend with sub-100ms harmonize-on-add, and cloud model serving for web, iOS, and browser-extension clients.

Founder

Mar 2026 – Present

LiminalML (liminalml.com)

Remote

- Solo-built and launched an AI-powered ML/SWE interview-prep SaaS: a Claude-backed tutor covers 157+ topics in resume-personalized, six-stage sessions with persisted, resumable state and inline math, code, and diagram rendering.
- Manage per-session LLM unit economics across free and paid tiers; built and run the full stack end-to-end (Next.js/TypeScript, Supabase, Stripe, PostHog).

Systems Engineering Intern

June 2023 – August 2024

Triaxis Power Consulting LLC

Charlotte, NC

- Scoped and deployed an IoT monitoring system (Raspberry Pi, thermocouples, potentiometers) tracking pipe temperature and thermal expansion in power plants for predictive maintenance.

PROJECTS

LexiMind: Multi-Task Post-Training Research | *cont. undergraduate independent research* Nov 2025 – Present

- Multi-task encoder-decoder (272M parameters) jointly trained on summarization, topic classification, and multi-label emotion detection.
- Shelved a completed working paper after an audit of the baselines and ablations showed the evaluation could not support its claims; published the postmortem and rebuilt the study around a question the evidence could answer.
- Running a fixed-budget comparison of multi-task post-training recipes (joint SFT, LoRA expert merging, frontier distillation, RL on verifiable rewards), with compute normalized in student optimizer tokens so the arms are genuinely comparable.
- Built the evaluation harness: pre-registered protocol, bootstrap confidence intervals, and per-class threshold calibration on a split disjoint from model selection.

EDUCATION

Appalachian State University

Boone, NC

B.S. Computer Science, Minor in Mathematics, Data Science Certificate

May 2026

- Senior Capstone (ML Lead): built the ML harmonization prototype adopted into production at the partnering startup, which became the foundation of the full-time founding-team role above.
- Relevant Coursework: Linear Algebra, Statistical Data Analysis, Numerical Methods, Advanced Reinforcement Learning, Applied Machine Learning, Theoretical Computer Science, Data Structures & Algorithms

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL

ML: PyTorch, Hugging Face Transformers, continued pre-training, SFT & RL post-training, distillation, GPU training, MLflow, Weights & Biases

Infrastructure: FastAPI, Docker, MongoDB, PostgreSQL/Supabase, Next.js/TypeScript, CI/CD